User's Manual

Model ZR22G, ZR402G Separate type Zirconia Oxygen Analyzer

Supplement

Thank you for selecting our Model ZR22G, ZR402G Separate type Zirconia Oxygen Analyzer.
In User's Manual, IM 11M12A01-02E, 6th Edition, supplied with this product, some corrections/additions have been made. Please replace the corresponding pages in your copy with the attached, revised pages.

Note

- p. 2-31, Section 2.7.8, " Pressure Regulator for Gas Cylinder (Part No. G7013XF or G7014XF)": Change drawing.
- p. 8-6, Section 8.2.3, "OutputHoud Setting": Changed value in Figure 8.2 and 8.3.
- p. 8-6, Section 8.2.4, "Default Value": Changed value and description in Table 8.2.
- p. 8-16, Section 8.6.1, "Setting the Date-and-Time": Added Item in Figure 8.13.
- p. 8-22, Section 8.6.4, "Setting Purging": Added Item.
- p. 8-23, Section 8.6.5, "Setting Passwords": Changed section number.
- p. 10-12, Section 10.3, "Operational Data Initalization": Changed value in Table 10.5.

2.7.7 Zero Gas Cylinder (Part No. G7001ZC)

The gas from this cylinder is used as the calibration zero gas and detector purge gas.

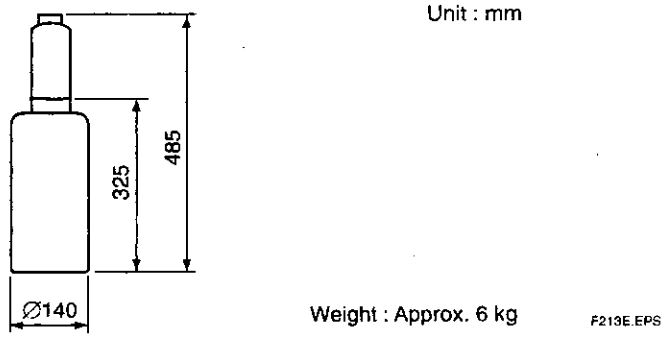
Standard Specifications

Capacity: 3.4 l

Filled pressure: 9.8 to 12 MPa G

Composition: 0.95 to 1.0 vol% O_2 in N_2

(Note) Export of such high pressure filled gas cylinder to most countries is prohibited or restricted.



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2.7.8 Cylinder Regulator Valve (Part No. G7013XF or G7014XF)

This regulator valve is used with the zero gas cylinders.

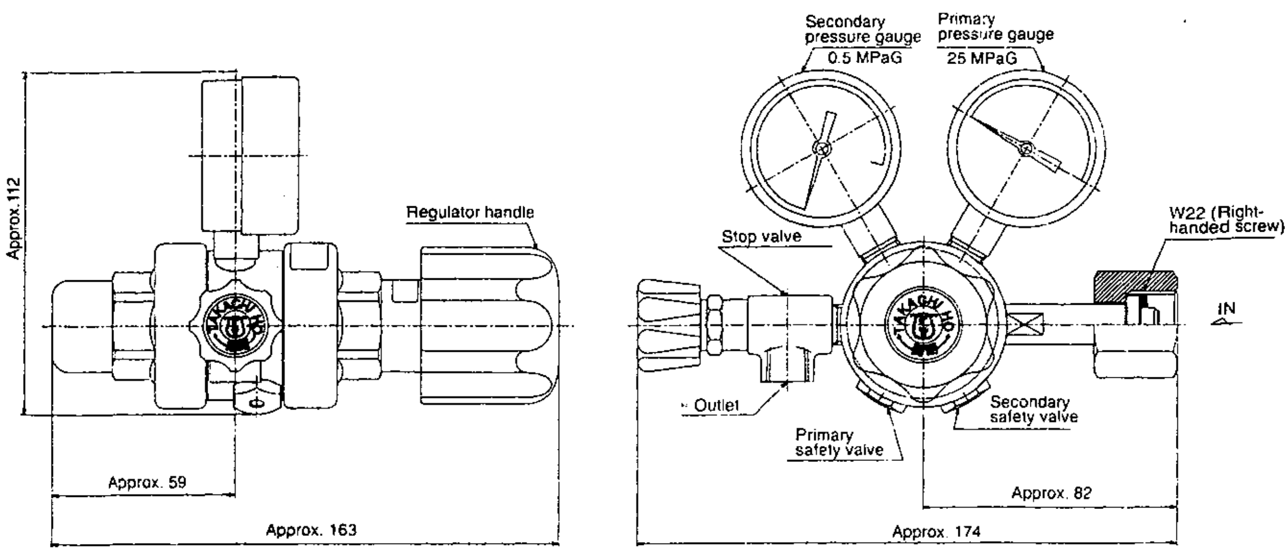
Standard Specifications

Pressure gauge: Primary 0 to 25 MPa G, Secondary 0 to 0.5 MPa G

Connection: Inlet W22 14 threads, right hand screw

Outlet Rc1/4 or 1/4FNPT

Material: Brass body



Part No.	* Outlet
G7013XF	Rc1/4
G7014XF	1/4 NPT female screw

8.2.3 Output Hold Setting

To set the output hold, follow these steps:

(1) Press the Setup key in the basic panel display to display the Execution/Setup display. Then select Setup in the Execution/Setup display. Next, select the mA-output setup and then the mA-output preset display as shown in Figure 8.2.

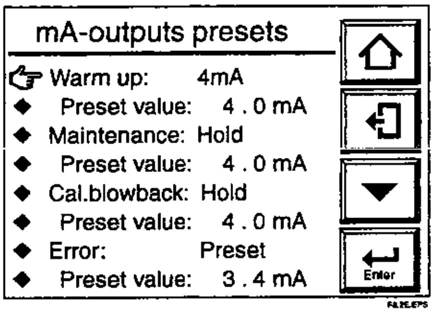


Figure 8.2 mA-output Preset Display

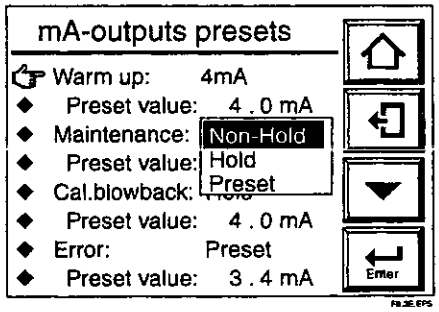


Figure 8.3

(2) From this display (Fig. 8.2), select the desired display. Figure 8.3 shows an example of selecting Maintenance. Select the desired output status.

(3) If a preset value is selected, set the corresponding output current. If you select a preset value just below Maintenance on the screen, the numeric-data entry display will appear. Enter the current value you want. To set 10 mA, type in 010 and press the [Enter] key to complete the setting. The setting range is from 2.4 to 21.6 mA.

8.2.4 Default Values

When the analyzer is delivered, or if data are initialized, output hold is the default as shown in Table 8.2.

Table 8.2 Output Hold Default Values

Status	Output hold (min. and max. values)
During warm-up	4 mA
Under maintenance	Holds output at value just before maintenance started.
Under calibration or blow-back	Holds output at value just before starting calibration or "blow-back."
On Error occurrence	Holds output at a preset value.
The default preset value is 3.4 mA only for output hold occurrence and others, it is 4 mA.	

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8.6 Other Settings

8.6.1 Setting the Date-and-Time

The following describe how to set the date-and-time. Automatic calibration or blowback works following this setting.

Proceed as follows:

(1) Press the Setup key in the Basic panel display to display the Execution/Setup display.

(2) Select Setup in the Execution/Setup display to display the "Commissioning" (Setup) display.

(3) Select Others in the "Commissioning" (Setup) display. The Others display then appears, as shown in Figure 8.14.

(4) Select Clock. The Clock display then appears, as shown in Figure 8.14.

(5) Select Set date to display the numeric-data entry display. To set the date June 21, 2000, enter 2106000 and press the [Enter] key. The display then returns to the one shown in Figure 8.14.

(5) Select Set time. Enter the time on a 24-hour basis. To enter 2:30 p.m., type in 1430 in the numeric-data entry display. Press the [Enter] key, and the time starts at 00 second.

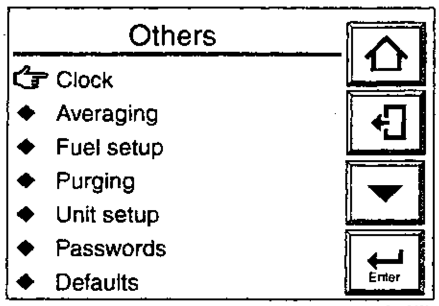

Figure 8.13 Other Settings

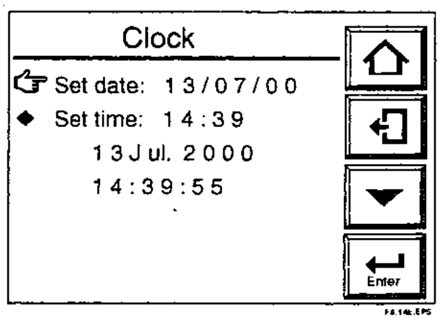

Figure 8.14 Clock Display

8.6.2 Setting Periods over which Average Values Are Calculated and Periods over which Maximum and Minimum Values Are Monitored

The equipment enables the display of oxygen concentration average values and maximum and minimum values under measurement (see Section 10.1.1, later in this manual). The following section describes how to set the periods over which oxygen concentration average values are calculated and maximum and minimum values are monitored.

8.6.2.1 Procedure

(1) Press the Setup key in the Basic panel display to display the Execution/Setup display.

(2) Select Setup in the Execution/Setup display to display the Commissioning" (Setup) display.

(3) Select Others in that display and then select Averaging in the Others display. The averaging display shown in Figure 8.15 then appears.

(4) Choose "Set period over which average is calculated" and enter the desired numeric value from the numeric-data entry display. To enter three hours, type in 003. The period over which average values can be calculated ranges from 1 to 255 hours.

(5) Choose "Set period over which maximum and minimum is stored" and enter the desired numeric value from the numeric-data entry display. To enter 48 hours, type in 048. The allowable input ranges from 1 to 255 hours.

8.6.4 Setting Purging

Purging is to remove condensed water in the calibration gas pipe by supplying a span calibration gas for a given length of time before warm-up of the detector. This prevents cell breakage during calibration due to condensed water in the pipe.

Open the solenoid valve for the automatic calibration span gas during purging and after the purge time has elapsed, close the valve to start warm-up.

Purging is enabled when the cell temperature is 100°C or below upon power up and the purge time is set in the range of 1 to 60 minutes.

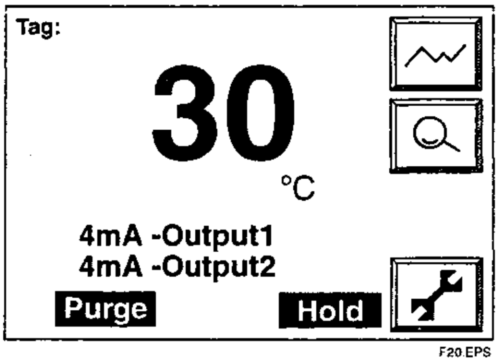

Figure 8.20 Display during Purging

8.6.4.1 Procedure

Set the purging time as follows:

(1) Press the Setup key in the basic panel display to display the Execution/Setup display.

(2) Select Setup in the Execution/Setup display. The "Commissioning" (Setup) display then appears.

(3) Select Others in that display and the Others display then appears, as shown in Figure 8.21.

(4) Select Purging. The purging time setting display appears, as shown in Figure 8.22.

(5) Point to the Purging time and press the [Enter] key. Then the display for selecting purging ime appears.

(6) Enter the desired numeric value from the numeric-data entry display.

The allowable input ranges from 0 to 60 minutes.

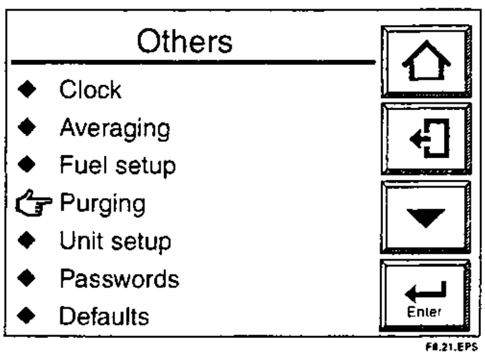

Figure 8.21 Other Settings

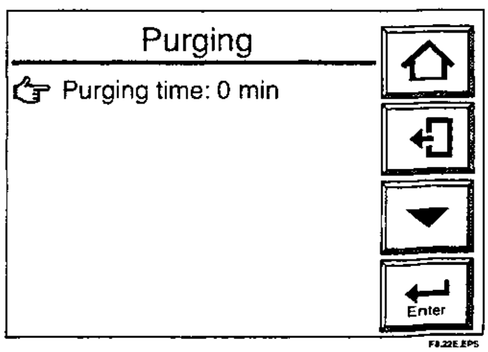

Figure 8.22 Purging Display

8.6.5 Setting Passwords

The converter enables password settings to prevent unauthorized switching from the Execution/Setup menu lower level menu displays. Set passwords for calibration, blowback and maintenance use and for setup use individually.

Proceed as follows:

(1) Press the Setup key in the basic panel display to display the Execution/Setup display.
(2) Choose Setup to display the "Commissioning" (Setup) display.
(3) Choose Others and then Passwords to display the Passwords display shown in Figure 8.20.
(4) Choose "Calibration, Blowback and Maintenance" to set passwords for calibration, blowback and maintenance respectively.
(5) The "text entry" display then appears. Enter up to eight alphanumeric characters as the password.
(6) In the same manner, follow steps 1 through 5 above to set a password for setup.
(7) Record passwords to manage them appropriately.

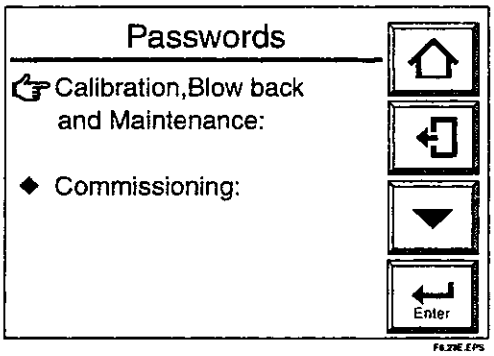

Figure 8.23 Passwords Display

If you forget a password, select Setup in the Execution/Setup display, and enter "MOON." By doing so, you can enter the Setup display only. Then display the Passwords and verify the set passwords.

Table 10.5 Initialization Items and Default Values

Item	Data to be initialized		Default setting
Equipment selection	Type of equipment		Not initialized
	Detector		ZR22
	Measurement gas		Wet gas
Displayed data	Display item	1st display item	Oxygen concentration
		2nd display item	Current output 1
		3rd display item	Current output 2
		Tag name	Deleted
	Trend graph	Parameter	Oxygen concentration
		Sampling interval	30 seconds
		Upper limit (graph)	25%O_2
		Lower limit (graph)	0%O_2
	Automatic return time		0 min.
	Language		Not initialized
Calibration data	Calibration setting	Mode	Manual
		Calibration procedure	Span - zero
		Zero-gas concentration	1.00%O_2
		Span-gas concentration	21.00%O_2
		Output hold time	10 min., 00 sec.
		Calibration time	10 min., 00 sec.
		Interval	30 days, 00 hr.
		Start date	01/01/00
		Start time	00:00
Blowback	Blowback setting	Mode	No function (invalid)
		(Output) hold time	10 min., 00 sec.
		Blowback time	10 min., 00 sec.
		Interval	30 days, 00 hr.
		Start date	01/01/00
		Start time	00:00
Current output data	mA-output 1 mA-output 2	Parameter	Oxygen concentration
		Min. oxygen concentration	0%O_2
		Max. oxygen concentration	25%O_2
		Output damping	0%
		Output mode	Linear
	Output hold setting	Warm-up	4 mA
		Set value	4 mA
		Maintenance	Previous value held
		Set value	4 mA
		Calibration, blowback	Previous value held
		Set value	4 mA
		Error	Preset value held
		Set value	3.4 mA

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